TaskTrack: A Proposed Console-Based Digital Task Organizer System

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Introduction to Computing Project
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1.1 Introduction

Task and deadline management is a critical aspect of academic success for tertiary students and educators, as it directly influences productivity, time management, and institutional workflow. According to Jia (2025), automated task scheduling and digital reminder systems play a significant role in reducing cognitive overload and preventing missed commitments. Furthermore, digital productivity tools provide structured frameworks that help learners and instructors systematically monitor academic responsibilities (Castro & Mohammad, 2024). For college students balancing multiple coursework demands and instructors managing assignments and project timelines, establishing a structured, centralized method for organizing tasks is essential for maintaining academic efficiency.

Within the setting of PHINMA Cagayan de Oro College, BSIT freshmen students and College of Information Technology Education Department instructors frequently manage academic tasks and deadlines across multiple subjects using informal, decentralized methods such as handwritten notes, smartphone memos, chat messages, or personal memory. However, when tasks are managed without a centralized or structured mechanism, three distinct problems occur. First, users frequently experience missed or forgotten deadlines because they lack automated tracking and reliable reminders to alert them of approaching due dates for assignments and submissions. Second, users face disorganized task records because academic and instructional details are scattered across disparate platforms, making information easily lost and time-consuming to retrieve. Third, users encounter difficulty in prioritizing tasks because they lack a single, sorted view of urgent versus completed responsibilities, resulting in rushed, last-minute work.

To address these challenges, the proponents developed TaskTrack: A Proposed Console-Based Digital Task Organizer System, a Python console-based application designed to streamline academic task management entirely through a command-line text interface. The system is designed specifically for BSIT freshmen students and College of Information Technology Education Department instructors at PHINMA Cagayan de Oro College. Operating via a text-based console menu, TaskTrack provides structured functions

to address the identified problems. Specifically, its deadline-tracking function alerts users to upcoming and overdue tasks to prevent missed deadlines; its task-management functions allow users to record, edit, search, and delete details in a single file-backed system using a plain text file to eliminate disorganized records; and its sorting and status-filtering features organize pending tasks by due date to resolve prioritization difficulties.

1.2 Statement of the Problem

General Problem

The current informal and manual process of tracking academic and instructional tasks among BSIT freshmen students and CITE Department instructors at PHINMA Cagayan de Oro College makes it difficult to organize assignments, monitor deadlines, and prioritize workload efficiently. Therefore, TaskTrack: A Proposed Console-Based Digital Task Organizer System is proposed to improve task organization, deadline tracking, and prioritization through a command-line environment.

Specific Problems

The proposed project specifically addresses the following problems:

1. Missed or Forgotten Deadlines – BSIT freshmen students and CITE Department instructors have no reliable automated reminders in the console, causing them to overlook due dates and submit incomplete or late coursework and evaluations.
2. Disorganized Task Records – Task and assignment details are scattered across informal notes and messages, resulting in lost information and wasted time searching for requirements.
3. Difficulty in Prioritizing Tasks – Without a centralized, sorted view of pending assignments and tasks in the text interface, users cannot identify urgent responsibilities, leading to last-minute, rushed work.

1.3 Objectives of the Study

General Objective

To develop TaskTrack: A Proposed Console-Based Digital Task Organizer System, a Python console-based application that enables BSIT freshmen students and CITE Department instructors at PHINMA Cagayan de Oro College to efficiently organize academic tasks, track deadlines, and prioritize coursework through a text-based console interface.

Specific Objectives

The project specifically aims to:

1. Display upcoming and overdue tasks through a console-based deadline-tracking function to address missed or forgotten deadlines.
2. Record, update, and manage task details in a plain text file (tasks.txt) via text menu prompts to resolve disorganized task records.
3. Sort and filter tasks by due date and completion status in the console display to address difficulty in prioritizing academic workload.

1.4 Scope

This project covers the development of TaskTrack: A Proposed Console-Based Digital Task Organizer System, a Python console-based application intended for BSIT freshmen students and CITE Department instructors at PHINMA Cagayan de Oro College. The system will operate entirely as a command-line program on a single computer console and will use a plain text file (tasks.txt) for persistent data storage.

The project includes the following:

- Task Management Console Function – Allows BSIT freshmen students and CITE Department instructors to add, edit, search, display, and delete academic tasks through text-based console prompts, saving all records directly to a plain text file.
- Deadline Tracking Console Function – Allows users to view categorized lists of upcoming and overdue tasks evaluated against the current system date.
- Sorting and Status Filtering Function – Allows users to mark tasks as completed and view pending tasks sorted chronologically by due date in the text display.

- Data Covered – Records, processes, displays, and stores task ID, title, course/subject, description, due date, priority level (High, Medium, Low), and completion status (Pending/Completed) within text file lines.

The project is limited to the processes and functions implemented by the proponents during the development period.

1.5 Limitations

The proposed system has the following limitations:

- The system is strictly console-based (command-line interface) and does not provide a graphical user interface (GUI) or web-based interface.
- The system is designed for BSIT freshmen students and CITE Department instructors at PHINMA Cagayan de Oro College and does not provide access for users outside the intended institutional setting.
- Data storage is limited to a local plain text file (tasks.txt), and the system does not use JSON, CSV, or a centralized online database.
- The system does not support simultaneous access by multiple users.
- The system does not provide real-time, push, email, or SMS notifications; deadline warnings are displayed only when the user launches the program and views the task display menu in the console.
- System security is limited to the input validation features implemented by the proponents; the system has no user login credentials or password protection.

1.6 System Requirements

Hardware Requirements

The system requires the following hardware:

- Computer or laptop with an Intel Core i3 processor or equivalent.
- At least 4 GB of RAM.
- At least 500 MB of available storage.
- Keyboard and monitor/display (for interacting with the console window).

Software Requirements

The system requires:

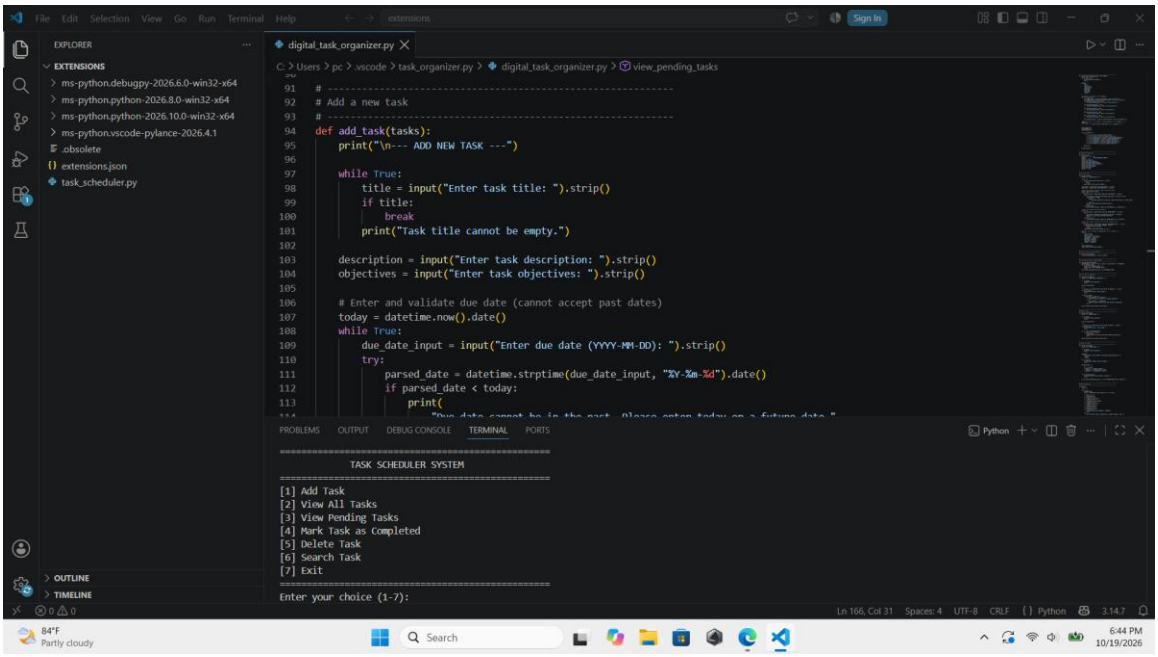
- Operating System: Windows 11 / Windows 10
- Python Version: Python 3.12
- Development Environment: Visual Studio Code
- Console Terminal: Command Prompt / Windows Terminal
- Required Libraries/Packages: Python standard library only (datetime)
- Storage: Plain text file (tasks.txt)

Functional Requirements

The system shall allow the BSIT freshmen students and CITE Department instructors to:

- Add a task in the console by entering the title, subject, description, priority level, and due date (YYYY-MM-DD), after which the system validates the inputs, appends the record to the tasks.txt file, and displays a confirmation message on the console.
- View upcoming and overdue tasks by selecting the tracking option from the menu, after which the system evaluates due dates against the system clock and displays formatted lists of urgent and past-due tasks in the text interface.
- Search, update, or delete existing task records in the console by entering the specific Task ID or title.
- Mark a task as completed by entering its Task ID, after which the system updates the task status to "Completed" and rewrites the updated records to the tasks.txt file.
- View tasks sorted by due date and completion status, displaying organized console tables of pending and finished assignments.
- Receive appropriate console validation messages when entering invalid menu choices, empty fields, or incorrectly formatted date strings.

1.7 Code Snippet of Scheduling System for Claiming School Uniform



Snippet 1 – Task Creation and Input Validation Function

Related Objective: Create and store tasks with complete and validated information.

Purpose: Demonstrates how the system accepts user input, validates task details, and stores a new task in the task list.

```
def add_task(tasks):
    print("\n--- ADD NEW TASK ---")

    while True:
        title = input("Enter task title: ").strip()
        if title:
            break
        print("Task title cannot be empty.")

    description = input("Enter task description: ").strip()
    objectives = input("Enter task objectives: ").strip()

    # Enter and validate due date (cannot accept past dates)
    today = datetime.now().date()
```

```
while True:
    due_date_input = input("Enter due date (YYYY-MM-DD): ").strip()
    try:
        parsed_date = datetime.strptime(due_date_input, "%Y-%m-%d").date()
        if parsed_date < today:
            print(
                "Due date cannot be in the past. Please enter today or a future date."
            )
            continue
        due_date = parsed_date.strftime("%Y-%m-%d")
        break
    except ValueError:
        print("Invalid format. Please use YYYY-MM-DD (e.g., 2025-12-31).")

# Enter and validate due time separately
while True:
    due_time_input = input("Enter due time (HH:MM AM/PM): ").strip()
    try:
        parsed_time = datetime.strptime(due_time_input, "%I:%M %p")
        due_time = parsed_time.strftime("%I:%M %p")
        break
    except ValueError:
        print("Invalid format. Please use HH:MM AM/PM (e.g., 02:30 PM).")

# Enter and validate priority (strictly High / Medium / Low)
while True:
    priority_input = input("Enter priority (High/Medium/Low): ").strip()
    if priority_input in ["High", "Medium", "Low"]:
        priority = priority_input
        break
    print("Please enter High, Medium, or Low.")
```

Explanation:

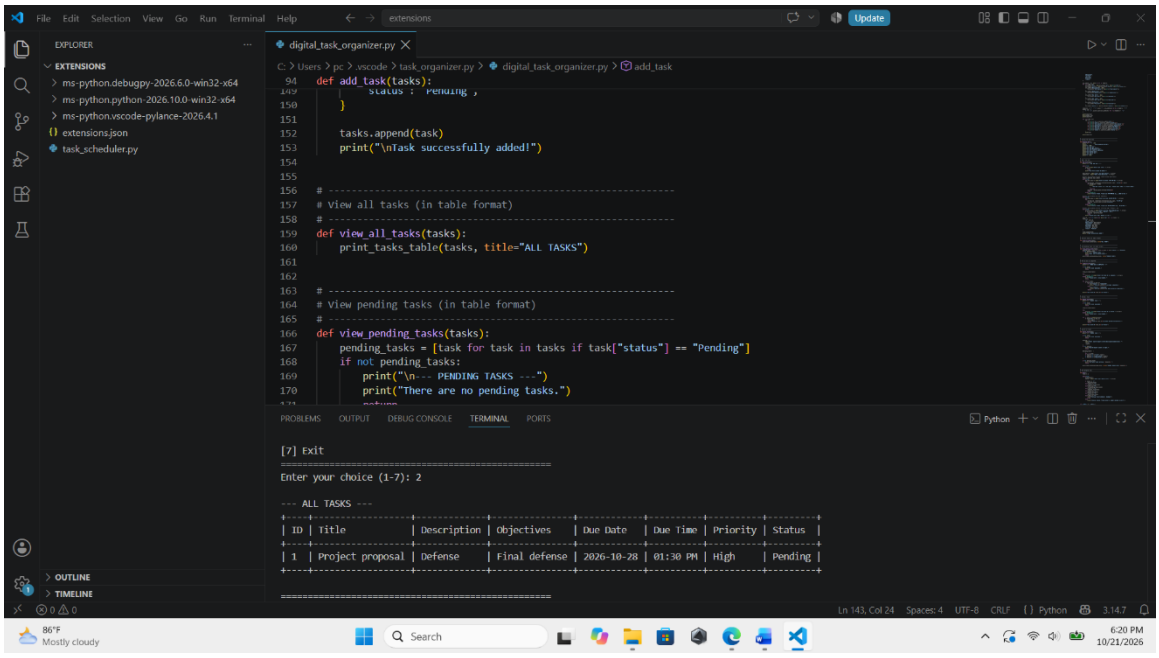
This function receives task information entered by the user, including the title, description, objectives, due date, and priority level. The function validates the input by ensuring that the task title is not empty, the due date is not in the past, and the priority value is either High, Medium, or Low.

After validation, the program automatically generates a unique task ID and creates a dictionary containing all task details. The task is then stored in the tasks list with a default status of Pending. This function supports Specific Objective 1 by allowing users to create and save organized task records accurately and efficiently.

Snippet 2 – View All Tasks Function

Related Objective: Display and organize all saved tasks for monitoring and tracking.

Purpose: Demonstrates how the system retrieves and presents all tasks in a structured table format.




```
def view_all_tasks(tasks):
    print_tasks_table(tasks, title="ALL TASKS")
```

Explanation:

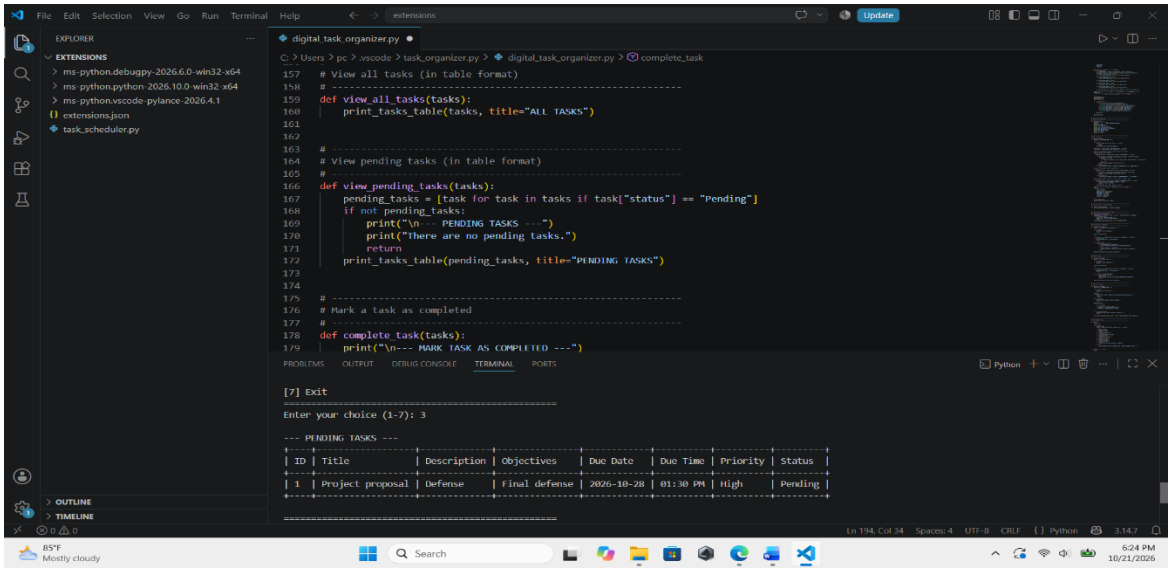
This function receives the list of stored tasks as input. It calls the print_tasks_table() function to process the task data and display it in a formatted table with columns such as ID, Title, Description, Objectives, Due Date, Due Time, Priority, and Status.

The function does not modify the stored data. Instead, it organizes and presents the information so users can easily review existing tasks. The output is a clear tabular display containing all recorded tasks. This function supports Specific Objective 2 by providing an organized view of task information, helping users monitor schedules and deadlines effectively.

Snippet 3 – View Pending Tasks Function

Related Objective: Filter and display tasks that have not yet been completed.

Purpose: Demonstrates how the system identifies pending tasks and displays only incomplete tasks.




```
def view_pending_tasks(tasks):
    pending_tasks = [task for task in tasks if task["status"] == "Pending"]
    if not pending_tasks:
        print("\n--- PENDING TASKS ---")
        print("There are no pending tasks.")
```

```
    return
print_tasks_table(pending_tasks, title="PENDING TASKS")
```

Explanation:

This function receives the complete task list and processes it using a list comprehension. It filters the tasks by checking whether the status field is equal to "Pending". Only tasks that have not yet been completed are included in the new list called pending_tasks.

If no pending tasks exist, the program displays a message informing the user. Otherwise, the filtered tasks are shown in a formatted table using the print_tasks_table() function. The output is a list containing only unfinished tasks. This function supports Specific Objective 3 by helping users focus on outstanding responsibilities and prioritize tasks that still require completion.

References

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- 2. Changyu Jia, Smart Scheduler and Reminder System. October 1, 2025. https://showcase.itcarlow.ie/C00292876/ResearchDocument-Scheduler&ReminderSystem.pdf